抗菌素级别概述

抗菌素，又称抗生素，是用于治疗细菌感染的关键药物[1]。它们通过杀死细菌或阻止其生长来发挥作用，是现代医学中不可或缺的一部分。然而，随着时间的推移，细菌已经进化出抵抗多种抗生素的能力，导致了抗生素耐药性这一日益严重的全球健康威胁[1]。为了有效应对这一挑战，并确保抗生素在未来仍然有效，深入理解抗生素的分类至关重要。对抗生素进行分类有助于医疗专业人员选择最合适的药物来治疗特定的感染，从而最大限度地提高治疗效果，同时减少不必要的广谱抗生素使用，以减缓耐药性的发展[7]。世界卫生组织（WHO）等机构也强调了对抗生素进行合理分类和管理的重要性，以应对日益严峻的耐药性问题[15]。本报告旨在全面概述抗菌素的分类方法、临床分级以及不同抗菌素类别的特点和应用场景。通过理解这些分类，医疗专业人员和相关领域的学生可以更深入地认识抗菌素的选择和使用策略[15]。

抗菌素的分类方法

抗菌素可以根据其多种关键特征进行分类[6]：

- 作用谱 **(Spectrum of Activity)**[24]：指抗菌素能够有效抑制或杀灭的细菌种类范围。
- 作用机制 **(Mechanism of Action)**[6]：描述抗菌素在细菌细胞或分子水平上发挥抑制或杀灭作用的具体方式。
- 化学结构 **(Chemical Structure)**[6]：根据抗菌素分子中存在的特定化学结构进行分组。
- 对细菌的影响 **(Effect on Bacteria)**[3]：区分抗菌素是直接杀死细菌（杀菌剂）还是仅抑制其生长（抑菌剂）。

这些分类方法并非完全互斥，一种抗菌素可以同时属于多个类别。例如，一种抗菌素可能具有特定的作用谱、通过特定的机制发挥作用、属于特定的化学结构类别，并且被归类为杀菌剂或抑菌剂。理解这些不同的分类角度有助于更全面地认识每种抗菌素的特性和临床应用[6]。

基于作用谱的分类

基于作用谱的分类是根据抗菌素能够有效靶向的细菌种类范围进行的[24]。这是初步选择抗菌素时的一个基本考虑因素。

- 窄谱抗菌素 **(Narrow-Spectrum Antibiotics)**
  - 定义：窄谱抗菌素仅对有限范围的细菌有效，通常只针对革兰氏阳性菌或革兰氏阴性菌中的一种[24]。例如，青霉素G主要针对革兰氏阳性菌[24]。
  - 机制：这类抗菌素通常靶向特定细菌群体独有的细菌结构或代谢途径。
  - 常见例子：青霉素G（主要针对革兰氏阳性菌）[24]，万古霉素（主要针对革兰氏阳性菌）[3]，非达霉素（特异性针对艰难梭菌）[34]，沙雷环素（针对特定的痤疮丙酸杆菌）[31]。
  - 临床意义：当已知致病菌时，窄谱抗菌素是首选，因为它们可以最大限度地减少对

正常菌群的干扰，并降低更广泛细菌产生耐药性的风险[26]。窄谱抗菌素的靶向作用有助于维持微生物群的平衡，与广谱抗菌素相比，可能减缓更广泛细菌产生耐药性的速度[26]。通过仅影响目标病原体，有益菌得以保留，从而降低了更广泛细菌种类产生耐药性的选择压力。

- 广谱抗菌素 **(Broad-Spectrum Antibiotics)**
  - 定义：广谱抗菌素对多种细菌有效，包括革兰氏阳性菌和革兰氏阴性菌[4]。例如，四环素和氯霉素对多种细菌都有效[25]。
  - 机制：这类抗菌素通常靶向更广泛细菌种类中保守的细菌结构或过程。
  - 常见例子：四环素类（如多西环素、米诺环素）[3]，氟喹诺酮类（如环丙沙星、左氧氟沙星）[3]，氯霉素[25]，氨基青霉素类（如阿莫西林、氨苄西林）[32]。
  - 临床意义：当致病菌未知或为多种细菌混合感染时，广谱抗菌素非常有用[3]。然而，不加选择地使用广谱抗菌素会显著促进抗生素耐药性的发展，并可能扰乱身体的正常菌群，导致继发感染[8]。虽然广谱抗菌素在病原体未知的情况下提供了初步治疗，但一旦确定病原体，就将治疗范围缩小到更窄谱的抗菌素，这是抗菌素管理的关键原则[49]。起始使用广谱抗菌素对于严重感染（病原体未知）可能挽救生命，但一旦培养结果出来，转换为更具针对性的窄谱药物可以最大限度地降低耐药性和对微生物群的破坏[49]。广谱抗菌素的便利性和感知疗效有时会导致其过度使用，这突显了需要更好的诊断工具和处方者教育，以促进有针对性的治疗[26]。在临床表现不明确提示特定细菌感染或缺乏快速诊断测试的情况下，倾向于开具广谱抗生素。然而，这种做法助长了日益严重的抗生素耐药性问题，并强调了准确诊断和有针对性治疗的重要性。
- 超广谱抗菌素 **(Extended-Spectrum Antibiotics)**
  - 定义：这些本质上是广谱抗菌素，但经过化学修饰后可以影响更多种类的细菌，通常包括可能对其他抗菌素产生耐药性的革兰氏阴性菌[24]。
  - 机制：超广谱抗菌素中的化学修饰通常针对细菌采用的特定耐药机制，或者与它们的母体化合物相比，扩大了易感细菌的种类范围。
  - 常见例子：一些更新代的头孢菌素，对更广泛的革兰氏阴性菌具有活性[38]，以及某些青霉素，如哌拉西林，对铜绿假单胞菌具有增强的活性[25]。
  - 临床意义：超广谱抗菌素在治疗对传统广谱抗菌素产生耐药性的细菌引起的感染方面发挥着关键作用，为日益严峻的感染提供了治疗选择。超广谱抗菌素的持续开发反映了对抗生素耐药性不断变化的努力，但这些药物最终也会出现耐药性，这需要持续的研究和新型抗菌策略的开发[38]。随着细菌进化并产生新的耐药机制，药物研究侧重于修改现有抗生素或发现具有更广泛活性或能够克服这些耐药性的新抗生素。这种持续的循环突显了对抗细菌感染斗争的动态性。

基于作用机制的分类

基于作用机制的分类是根据抗菌素靶向的特定分子靶点以及它们如何抑制或杀死细菌来进行的[6]。理解作用机制对于预测抗菌素的有效性和潜在的副作用至关重要。

- 细胞壁合成抑制剂 **(Inhibitors of Cell Wall Synthesis)**
  - 机制：这类抗菌素干扰细菌细胞壁中肽聚糖的合成。通过抑制这个过程，它们削弱细胞壁，导致细菌裂解和死亡。这种机制通常使其成为杀菌剂 [3]。
  - 常见例子：
    - **β-**内酰胺类 **(Beta-Lactams)**：这是一个庞大的类别，其特征是存在β-内酰胺环，包括青霉素类（如阿莫西林、青霉素G）[3]，头孢菌素类（如头孢曲松、头孢氨苄）[3]，碳青霉烯类（如美罗培南、亚胺培南）[3]，和单环β-内酰胺类（如氨曲南）[21]。
    - 糖肽类 **(Glycopeptides)**：包括万古霉素和替考拉宁，它们干扰肽聚糖的交联 [3]。
    - 肽类 **(Peptides)**：杆菌肽抑制肽聚糖前体的转运 [22]。
      - β-内酰胺类药物虽然具有抑制细胞壁合成的核心机制，但其对不同细菌种类的有效性以及对β-内酰胺酶等细菌酶的敏感性存在差异。这促使了β-内酰胺酶抑制剂的开发，以增强其活性 [6]。虽然基本作用相同，但β-内酰胺环内的结构差异以及特定侧链的存在与否会影响药物与不同细菌的青霉素结合蛋白（PBPs）的结合能力及其对酶降解的抵抗力。
      - 糖肽类药物，特别是万古霉素，对于治疗对β-内酰胺类药物耐药的革兰氏阳性菌（如耐甲氧西林金黄色葡萄球菌，MRSA）引起的感染至关重要。它们独特的作用机制，靶向细胞壁合成的不同步骤，使其成为重要的替代选择 [3]。糖肽类药物与肽聚糖前体的D-Ala-D-Ala末端结合，从而阻止交联，这种作用机制不受β-内酰胺酶的影响，使其对许多耐药菌株有效。
- 蛋白质合成抑制剂 **(Inhibitors of Protein Synthesis)**
  - 机制：这类抗菌素通过靶向细菌核糖体的不同组分（30S或50S亚基）来干扰细菌蛋白质合成的复杂过程。这种干扰阻止了细菌产生生长和生存所需的必需蛋白质 [6]。此类药物包括抑菌剂（如四环素类、大环内酯类）和杀菌剂（如高浓度时氨基糖苷类）[3]。
  - 常见例子：
    - 大环内酯类 **(Macrolides)**：如红霉素、阿奇霉素、克拉霉素，与50S核糖体亚基结合，阻止肽链延伸 [3]。
    - 四环素类 **(Tetracyclines)**：如四环素、多西环素、米诺环素，通过阻止tRNA与30S核糖体结合来抑制细菌蛋白质合成 [3]。
    - 氨基糖苷类 **(Aminoglycosides)**：如庆大霉素、妥布霉素、阿米卡星，不可逆地与30S核糖体亚基结合，导致mRNA误读和蛋白质合成抑制 [3]。
    - 林可霉素类 **(Lincosamides)**：如克林霉素、林可霉素，与50S核糖体亚基结合并抑制蛋白质合成 [3]。
    - 噁唑烷酮类 **(Oxazolidinones)**：如利奈唑胺、替地唑胺，通过与50S核糖体亚基结合来抑制蛋白质合成的起始阶段 [3]。
    - 氯霉素 **(Chloramphenicol)**：与50S核糖体亚基结合并阻止肽链延伸 [21]。
    - 链霉素类 **(Streptogramins)**：如奎奴普汀/达福普汀，协同作用抑制50S核糖

体亚基的蛋白质合成[6]。
    - 甘氨酰环胺类 **(Glycylcyclines)**：如替加环素，与四环素类相关，通过与30S核糖体亚基结合来抑制蛋白质合成[3]。
        - 蛋白质合成抑制剂种类繁多，每种都靶向细菌核糖体的特定阶段或亚基，这导致了它们在作用谱、效力和副作用方面的差异。这也意味着耐药机制可能特定于每个类别甚至类别内的个别药物[27]。
        - 一些蛋白质合成抑制剂，如大环内酯类和四环素类，不仅具有抗菌活性，还表现出抗炎和免疫调节作用。这种双重作用在某些临床情况下可能是有益的，但也增加了其药理学复杂性和潜在的副作用[104]。
        - 氨基糖苷类药物虽然对许多需氧革兰氏阴性菌感染非常有效，但与明显的毒性相关，特别是耳毒性和肾毒性。这需要仔细监测接受这些药物治疗的患者，并在可能的情况下考虑替代药物[3]。
- 核酸合成抑制剂 **(Inhibitors of Nucleic Acid Synthesis)**
    - 机制：这类抗菌素干扰细菌DNA复制、修复或RNA转录等基本过程。通过破坏这些过程，它们阻止细菌繁殖，并可能导致细胞死亡[6]。
    - 常见例子：
        - 氟喹诺酮类 **(Fluoroquinolones)**：如环丙沙星、左氧氟沙星、莫西沙星，抑制细菌DNA促旋酶和拓扑异构酶，这些酶对于DNA复制和修复至关重要[3]。
        - 利福霉素类 **(Rifamycins)**：如利福平、利福布汀、利福昔明，抑制细菌DNA依赖的RNA聚合酶，从而阻止RNA合成[6]。
        - 硝基咪唑类 **(Nitroimidazoles)**：如甲硝唑，产生代谢副产物，破坏细菌DNA，尤其是在厌氧菌中[3]。
            - 氟喹诺酮类药物虽然对多种细菌有效，但与严重的副作用相关，包括肌腱断裂、心脏问题和神经系统问题，导致对其使用发出警告和限制。这突显了在处方前仔细权衡风险和益处的重要性[3]。氟喹诺酮类药物抑制细菌DNA拓扑异构酶的机制也可能对哺乳动物酶产生脱靶效应，从而导致观察到的毒性。
            - 利福霉素类药物由于其能够特异性抑制细菌RNA聚合酶，因此在治疗分枝杆菌感染（特别是结核病和麻风病）方面发挥着基石作用。它们对快速分裂和休眠细菌都有效，这使其在长期治疗方案中至关重要[6]。细菌RNA聚合酶与哺乳动物RNA聚合酶的独特结构差异使得利福霉素类药物能够选择性地靶向和抑制细菌转录，这是细菌生存的关键过程。
- 代谢途径抑制剂 **(Inhibitors of Metabolic Pathways)**
    - 机制：这类抗菌素干扰细菌的基本代谢途径，通常通过充当抗代谢物，类似于天然底物并竞争性地抑制关键酶。一个常见的例子是抑制叶酸合成，叶酸对于核酸和某些氨基酸的产生至关重要[6]。
    - 常见例子：
        - 磺胺类 **(Sulfonamides)**：如磺胺甲噁唑（常与甲氧苄啶合用），竞争性地抑制叶酸合成途径中的二氢蝶酸合成酶[3]。

        - 甲氧苄啶 **(Trimethoprim)**：抑制叶酸合成途径中的二氢叶酸还原酶，常与磺胺类药物协同使用 [23]。
            - 磺胺类药物的选择性毒性源于细菌自身合成叶酸，而人类从饮食中获取。这种差异使得磺胺类药物能够抑制细菌生长，而不会显著影响人类代谢。然而，细菌对磺胺类药物的耐药性非常普遍 [3]。通过靶向细菌生存所必需的特定代谢途径，并利用细菌和人类在叶酸获取方面的差异，磺胺类药物提供了一个有价值的治疗选择。然而，这些药物的广泛使用导致了耐药菌株的选择。
- 细胞膜功能破坏剂 **(Disruptors of Cell Membrane Function)**
    - 机制：这类抗菌素与细菌细胞膜相互作用，破坏其完整性并增加其通透性。这导致必需的细胞内成分泄漏，最终导致细菌死亡 [25]。
    - 常见例子：
        - 多粘菌素类 **(Polymyxins)**：如多粘菌素B、粘菌素，与细菌细胞膜中的磷脂相互作用，破坏其结构和功能 [18]。
        - 环状脂肽类 **(Cyclic Lipopeptides)**：如达托霉素，插入细菌细胞膜，引起快速去极化和细胞死亡 [3]。
            - 多粘菌素类药物通常作为治疗由多重耐药革兰氏阴性菌（如铜绿假单胞菌和碳青霉烯耐药肠杆菌科细菌）引起的严重感染的最后手段，因为它们具有潜在的肾毒性和神经毒性 [18]。多粘菌素类药物独特的作用机制，靶向细菌细胞膜，在其他抗生素因耐药性无效时提供了一个有价值的选择。然而，它们的毒性限制了其更广泛的应用。

基于化学结构的分类

抗菌素也根据其共享的化学结构进行分类，因为同一结构类别的药物通常表现出相似的作用机制、抗菌活性和潜在的过敏反应 [6]。

- **β-**内酰胺类抗生素 **(Beta-Lactam Antibiotics)**：以其结构中存在的四元β-内酰胺环为特征 [6]。
    - 青霉素类 **(Penicillins)**：如阿莫西林、氨苄西林、青霉素G、哌拉西林，是β-内酰胺类的一个亚类，常用于治疗多种感染，特别是革兰氏阳性菌和一些革兰氏阴性球菌引起的感染 [3]。
    - 头孢菌素类 **(Cephalosporins)**：如头孢氨苄、头孢唑林、头孢曲松、头孢吡肟，是β-内酰胺类的另一个主要亚类，通常按代分类，后代药物对革兰氏阴性菌的活性增强 [3]。
    - 碳青霉烯类 **(Carbapenems)**：如亚胺培南、美罗培南、厄他培南，是一类超广谱β-内酰胺类抗生素，常用于严重感染或由多重耐药菌引起的感染 [3]。
    - 单环**β-**内酰胺类 **(Monobactams)**：如氨曲南，是一种具有单环结构的β-内酰胺类抗生素，通常对需氧革兰氏阴性菌有效 [21]。
        - 共享的β-内酰胺环结构是其共同作用机制（抑制细胞壁合成）的基础，也是该

类别内过敏反应发生交叉反应的潜在原因[23]。β-内酰胺环可能被免疫系统识别为过敏原，对一种β-内酰胺类抗生素过敏的个体由于这种共享的结构基序，对其他药物过敏的风险也会增加。

- 大环内酯类抗生素 **(Macrolides)**：如红霉素、克拉霉素、阿奇霉素，以其庞大的大环内酯环为特征，主要抑制细菌蛋白质合成[3]。
- 四环素类抗生素 **(Tetracyclines)**：如四环素、多西环素、米诺环素，具有特征性的四环化学结构，并抑制细菌蛋白质合成[3]。
- 氨基糖苷类抗生素 **(Aminoglycosides)**：如庆大霉素、妥布霉素、阿米卡星，含有通过糖苷键连接的氨基糖，并抑制细菌蛋白质合成[3]。
- 氟喹诺酮类抗生素 **(Fluoroquinolones)**：如环丙沙星、左氧氟沙星、莫西沙星，具有氟化的喹诺酮核心结构，并抑制细菌DNA合成[3]。
- 糖肽类抗生素 **(Glycopeptides)**：如万古霉素、替考拉宁，以糖基化的环状或多环非核糖体肽为特征，并抑制细菌细胞壁合成[3]。
- 林可霉素类抗生素 **(Lincosamides)**：如克林霉素、林可霉素，具有特定的化学结构，并抑制细菌蛋白质合成[3]。
- 噁唑烷酮类抗生素 **(Oxazolidinones)**：如利奈唑胺、替地唑胺，含有特征性的噁唑烷酮环，并抑制细菌蛋白质合成的起始阶段[3]。
- 利福霉素类抗生素 **(Rifamycins)**：如利福平、利福布汀、利福昔明，是安沙霉素类的一个亚类，以其复杂的巨环结构为特征，并抑制细菌RNA聚合酶[6]。
- 磺胺类抗生素 **(Sulfonamides)**：如磺胺甲噁唑，含有一个磺酰胺化学基团，并抑制细菌叶酸合成[3]。
- 多粘菌素类抗生素 **(Polymyxins)**：如多粘菌素B、粘菌素，是多肽类抗生素，可破坏细菌细胞膜功能[18]。

抗菌素的临床分级和管理

抗菌素的分类在临床实践中对于指导治疗选择起着至关重要的作用。医疗专业人员根据感染的类型、最可能的致病菌及其药敏谱来选择合适的抗菌素[3]。抗菌素管理计划在医疗机构中发挥着越来越重要的作用，旨在促进抗菌素的合理使用，改善患者预后，减少抗生素耐药性，并降低多重耐药菌感染的传播[7]。这些计划通常包括实施指南、监测抗菌素使用情况以及对医疗专业人员和患者进行教育。

世界卫生组织（WHO）的AWaRe（可获得性、观察性、储备性）分类是全球抗菌素管理的关键工具[15]。

- 可获得性类 **(Access Group)**：包括治疗常见感染的一线和二线抗菌素，这些药物的耐药性风险较低，如阿莫西林、苄青霉素、头孢氨苄、甲硝唑[16]。这些药物应广泛可用，并用于治疗大部分感染。
- 观察性类 **(Watch Group)**：包括具有较高耐药性风险的抗菌素，通常是广谱药物，如环丙沙星、克拉霉素和万古霉素[16]。其使用应受到仔细监测，并仅限于特定适应症。

- 储备性类 **(Reserve Group)**：包括最后手段的抗菌素，如粘菌素和利奈唑胺 [16]，仅用于其他选择失败时，治疗已确诊或疑似多重耐药菌感染。

世界卫生组织的AWaRe分类系统提供了一个全球认可的框架，根据抗菌素在人类医学中的重要性和耐药性发展的风险对其进行分类。该系统旨在指导抗菌素处方实践，并通过鼓励优先使用“可获得性”组中的抗菌素，并将“观察性”和“储备性”组中的抗菌素用于特定的有限情况，从而促进全球的抗菌素管理工作 [15]。通过将抗菌素分为这三类，世界卫生组织旨在改善基本抗菌素的可及性，同时通过促进对那些具有较高耐药性倾向的抗菌素的负责任使用，来减轻抗生素耐药性的威胁。

临床抗菌素使用指南通常针对特定感染进行调整，并考虑到当地的抗菌素耐药性模式。这突显了使用当地药敏试验结果和循证指南来指导抗菌素选择的重要性，而不是仅仅依赖于广泛的分类 [11]。虽然广泛的分类提供了一般的理解，但对于特定地理位置的特定感染，最佳的抗菌素选择取决于当地细菌群体中不同药物的耐药性流行程度。因此，当地的药敏数据对于有效的经验性和确定性治疗至关重要。

抗菌素特性和应用总结

下表总结了报告中讨论的不同抗菌素类别。

抗菌素类别	化学结构类别	主要作用机制	作用谱	杀菌/抑菌	常见例子	典型临床应用	**WHO AWaRe** 类别
青霉素类	β-内酰胺类	抑制细胞壁合成	窄谱（主要针对革兰氏阳性菌）	杀菌	阿莫西林、青霉素G	链球菌感染、梅毒	可获得性
头孢菌素类	β-内酰胺类	抑制细胞壁合成	广谱（不同代作用谱有所不同）	杀菌	头孢曲松、头孢氨苄	肺炎、尿路感染、脑膜炎	观察性
碳青霉烯类	β-内酰胺类	抑制细胞壁合成	超广谱	杀菌	亚胺培南、美罗培南	严重感染、多重耐药菌感染	储备性
单环β-内酰胺类	β-内酰胺类	抑制细胞壁合成	窄谱（主要针对需氧革兰氏	杀菌	氨曲南	医院获得性感染	观察性

			阴性菌）				
大环内酯类	大环内酯类	抑制蛋白质合成	广谱	抑菌（高浓度时可能杀菌）	红霉素、阿奇霉素	呼吸道感染、性传播感染	观察性
四环素类	四环素类	抑制蛋白质合成	广谱	抑菌	多西环素、米诺环素	非典型肺炎、痤疮	可获得性/观察性
氨基糖苷类	氨基糖苷类	抑制蛋白质合成	广谱（主要针对革兰氏阴性菌）	杀菌	庆大霉素、阿米卡星	严重革兰氏阴性菌感染	观察性/储备性
林可霉素类	林可酰胺类	抑制蛋白质合成	窄谱（主要针对革兰氏阳性菌和厌氧菌）	抑菌/杀菌（取决于浓度）	克林霉素	皮肤软组织感染、厌氧菌感染	可获得性
噁唑烷酮类	噁唑烷酮类	抑制蛋白质合成	窄谱（主要针对革兰氏阳性菌）	抑菌	利奈唑胺、替地唑胺	耐药革兰氏阳性菌感染	储备性
利福霉素类	安沙霉素类	抑制RNA合成	广谱（对分枝杆菌尤其有效）	杀菌	利福平、利福昔明	结核病、麻风病	可获得性/观察性/储备性
磺胺类	磺胺类	抑制叶酸合成	广谱	抑菌	磺胺甲噁唑	尿路感染、诺卡菌病	可获得性
多粘菌素类	多肽类	破坏细胞膜功能	窄谱（仅对革兰氏阴性菌有效）	杀菌	多粘菌素B、粘菌素	多重耐药革兰氏阴性菌感染	储备性
糖肽类	糖肽类	抑制细胞壁合成	窄谱（仅对革兰氏阳性菌有	杀菌	万古霉素、替考拉宁	耐药革兰氏阳性菌感染	观察性/储备性

			效）				
甘氨酰环胺类	甘氨酰环胺类	抑制蛋白质合成	广谱	抑菌	替加环素	复杂性腹腔感染、皮肤软组织感染	观察性
硝基咪唑类	硝基咪唑类	破坏DNA结构	窄谱（主要针对厌氧菌和某些寄生虫）	杀菌	甲硝唑	厌氧菌感染	可获得性
环状脂肽类	脂肽类	破坏细胞膜功能	窄谱（仅对革兰氏阳性菌有效）	杀菌	达托霉素	耐药革兰氏阳性菌感染	储备性

结论

全面而细致地理解抗菌素的分类，包括其作用谱、作用机制和化学结构，对于临床医生做出明智的处方决策至关重要。这种知识不仅对于实现最佳的患者治疗效果至关重要，而且在全球范围内通过促进负责任的使用和管理，对抗生素耐药性的持续威胁方面也发挥着至关重要的作用[7]。通过整合对这些不同分类系统的认识，临床医生可以选择最适合特定感染的抗菌素，从而最大限度地降低耐药性发展的风险，并确保为患者提供最佳的治疗。

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